Workshop Feature Extraction/Selection:

Datasets

Use one dataset from the ten high-dimensional microarray datasets download from UCI REPOSITORY (https://archive.ics.uci.edu/).

They usually comprise several thousands of features. Table 1 describes the detailed information of the data sets.

Replace the missing value existed in lymphoma with the mean of values (you can use an other preparation/cleaning method you learned).

You can download another dataset with few samples and thousands of features.

Table 1: Datasets details.

Datasets	Number of features	Number of instances	Number of classes
Colon tumor	2000	62	2
Central Nervous System	7129	60	2
Leukemia	7129	72	2
Breast cancer	24481	97	2
Lung_cancer	12533	181	2
Ovarian cancer	15154	253	2
Leukemia 3 classes	7129	72	3
Luekemia 4 classes	7129	72	4
Lymphoma	4026	62	3
MLL	12582	72	3

1- Study classification results obtained when using no feature extraction methods, then the results achieved by applying studied feature extraction method (s).
2- Study classification results obtained when the results achieved by applying different filter feature selection methods (and later Wrapper and Embedded feature selection methods).
3- Compare obtained classification results with feature selection methods and extraction feature methods. What are your conclusions....

NB/ you can use three or more studied classification methods or other classification methods you know.